

**11 (a)** frequency of carrier wave varies (in synchrony) with signal  
(in synchrony) with displacement of signal

M1  
A1 [2]

**(b)** advantages e.g. less noise / less interference  
greater bandwidth / better quality  
(1 each, max 2)  
disadvantages e.g. short range / more transmitters / line of sight  
more complex circuitry  
greater expense  
(1 each, max 2)

B4 [4]

**12 (a)** gain / loss/dB =  $10 \lg(P_1/P_2)$

C1